



UNSW
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

**EDUCATIONAL ASSESSMENT
AUSTRALIA**



ICAS

**INTERNATIONAL COMPETITIONS AND
ASSESSMENTS FOR SCHOOLS**

MATHEMATICS 2009

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL and OTHER INFORMATION.**

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET.**

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

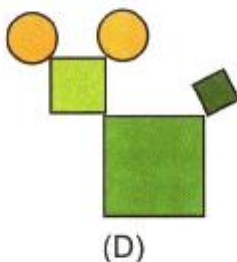
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

**PLEASE SEE BACK COVER FOR A LIST
OF THE YEAR LEVELS THAT SHOULD
SIT THIS PAPER**

1. Which picture is made from two circles and four squares?



2. Nina works every day except Saturdays and Sundays.

February						
S	M	T	W	T	F	S
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28			

How many days in February does she work?

- (A) 8
(B) 20
(C) 24
(D) 28

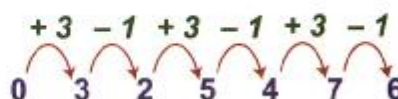
3. Brian has this picture.



Which of these shows Brian's picture after a quarter turn?



4. The picture shows how Simon made a number pattern.



What number comes next in Simon's pattern?

- (A) 3
(B) 5
(C) 7
(D) 9

5. Bron thinks of a number where:

- the thousands digit is even
- the hundreds digit is even
- the tens digit is odd.

Which of these could be her number?

- (A) 4671
(B) 5743
(C) 6532
(D) 9264

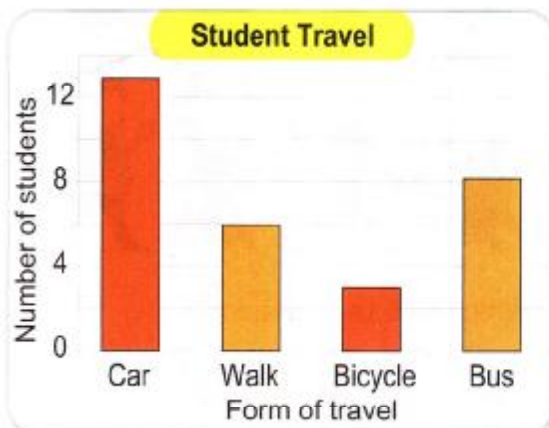
6. In this number sentence, **?** has the same value on both sides.

$$35 + \text{?} = 6 \times \text{?}$$

What is the missing number?

- (A) 4
(B) 5
(C) 6
(D) 7

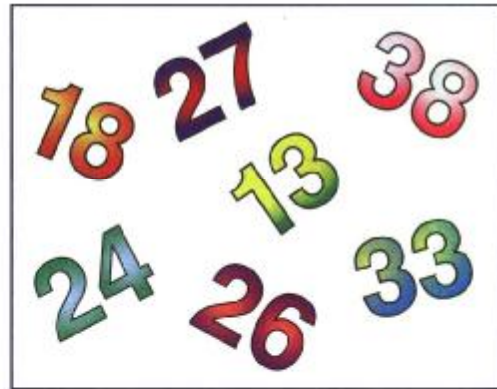
7. Ed surveyed his class to find out how students travel to school.



How many students in Ed's class walk to school?

- (A) 13
(B) 8
(C) 6
(D) 5

8.



How many of these numbers have no remainder when divided by three?

- (A) 7
(B) 4
(C) 3
(D) 2

9. $1091 - 608 = \text{?}$

What is the missing number?

- (A) 483
(B) 497
(C) 683
(D) 697

10. Julia wants to tile this area of her bathroom wall.



Which of these tiles can she use to cover this area with no cutting, overlapping or gaps?



- (A) (B) (C) (D)

11. A stick is painted with equal strips of red and white.



Which picture shows half of this stick?



(A)



(B)



(C)



(D)

12. $79 + 52 + 88 = ?$

- (A) 209
(B) 219
(C) 2019
(D) 2119

13. Daniel's watering can holds 6 L of water when full. He fills his watering can 3 times from the bucket and then the bucket is empty.



How much **more** water does one full bucket hold than Daniel's full watering can?

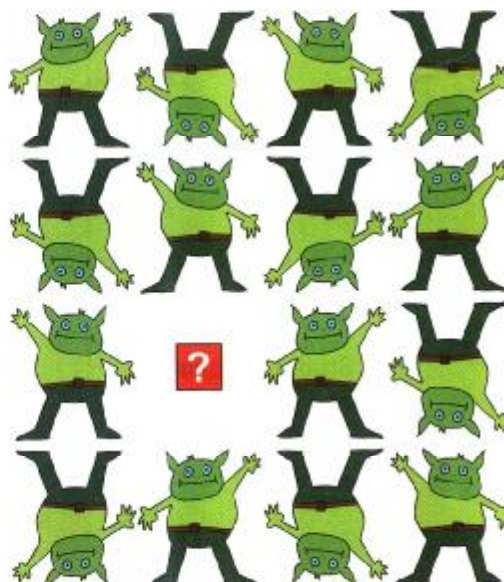
- (A) 3 L
(B) 9 L
(C) 12 L
(D) 18 L

14. How many of these shapes have at least one right angle?

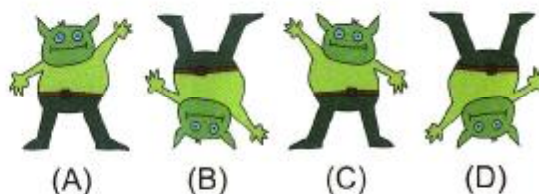


- (A) 2
(B) 3
(C) 4
(D) 5

15. A picture is missing from this pattern.



Which picture is missing?



16. Sasi has these cards.



She picks one card without looking.

Which card is she most likely to pick?



(A)



(B)



(C)



(D)

17. Eva wrote down some facts about two numbers using symbols.

$$\text{♣} + \text{♠} = 5$$

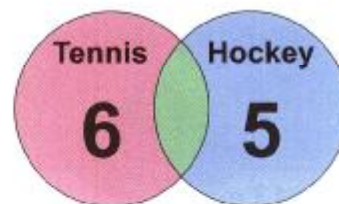
$$\text{♣} \times \text{♠} = 6$$

$$\text{♣} - \text{♠} = 1$$

Which numbers do the symbols stand for?

	♣	♠
(A)	3	2
(B)	4	1
(C)	5	4
(D)	6	0

18. In a group of 20 students there were some who played hockey only, some who played tennis only and some who played both.



How many students played **both** tennis and hockey?

- (A) 9
(B) 11
(C) 14
(D) 15

19. Kavi watched a movie. He looked at a clock when the movie started and when the movie finished.



Movie started

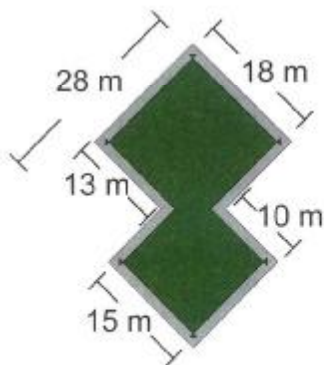


Movie finished

How many minutes long was the movie?

- (A) 120
(B) 130
(C) 135
(D) 210

20. This diagram shows a fence around a garden. Some of the measurements are not shown.



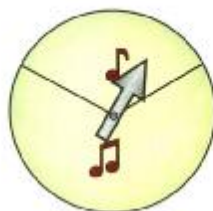
What is the length of the fence?

- (A) 56 m
- (B) 84 m
- (C) 112 m
- (D) 140 m

21. Which spinner has the least chance of landing on  ?



(A)



(B)



(C)



(D)

22. Jake drew this picture of himself.



Which of these is a mirror image of Jake's picture?



(A)



(B)

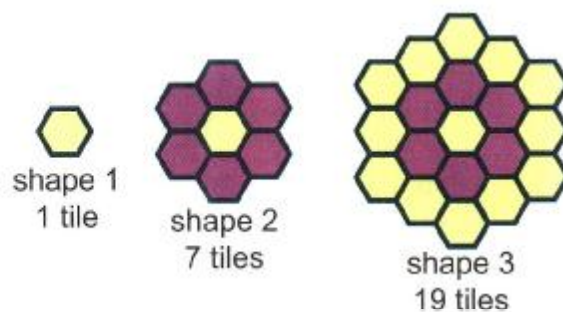


(C)



(D)

23. Arvind is making a pattern of shapes out of tiles.



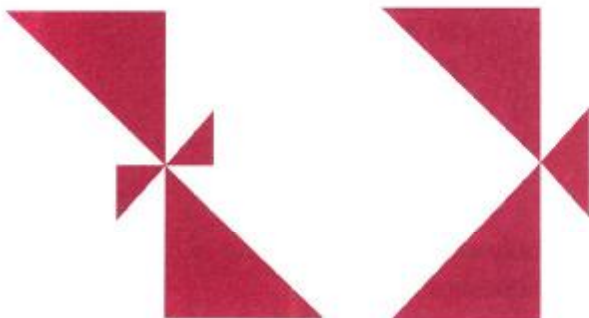
How many tiles will be in shape 4?

- (A) 18
- (B) 31
- (C) 37
- (D) 43

24. Hayley and Ravi are sitting at opposite ends of a rectangular table.

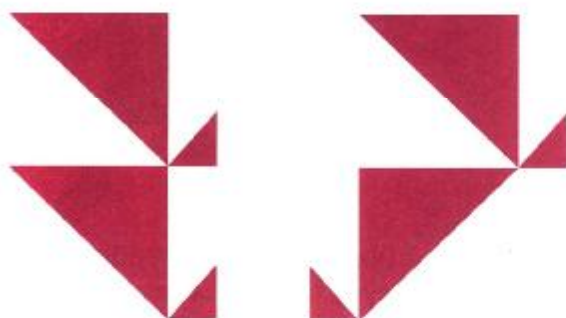
Hayley places some shapes on the table.

Which of these shapes would look the same to Ravi?



(A)

(B)



(C)

(D)

25. Lena saw this sign as she began her trip.

Pelican Point	35 km
Tern Village	105 km
Gulls Town	175 km

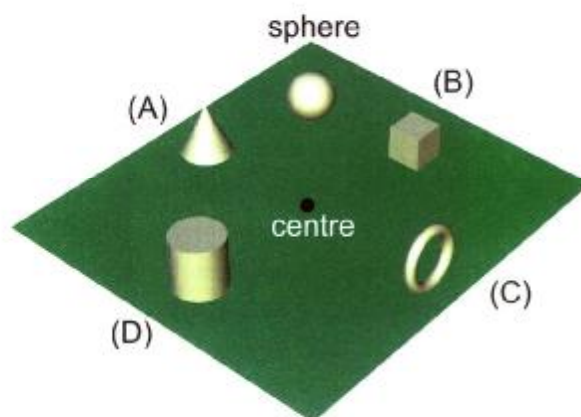
When she arrived at Tern Village, she drove back to Pelican Point. She then drove to Gulls Town.

What is the total distance Lena drove?

- (A) 210 km
(B) 280 km
(C) 315 km
(D) 385 km

26. The picture shows five sculptures in a park. The sphere is south-east of the centre.

Which sculpture is north of the centre?



27. Sean had 24 coloured pencils.

He was given a new set of coloured pencils which contained twice as many as he already had.

Sean gave 11 coloured pencils to each of his three sisters.

How many pencils did he now have?

- (A) 36
- (B) 37
- (C) 39
- (D) 61

28. Here is a bottle of medicine.



Which of these shows how to work out how many doses there are in the bottle?

- (A) $360 \div 8$
- (B) $360 \div 3$
- (C) 8×360
- (D) 3×8

29. Ivan has 6 pieces of string. Each piece of string is 25 cm. He ties the strings together to make one string that is 90 cm long. He uses the same amount of string for each knot.

How much string does he use in one knot?

- (A) 10 cm
- (B) 12 cm
- (C) 15 cm
- (D) 18 cm

30. Sally wrote a story about two kinds of monsters.

Sally's monsters



1 head
3 eyes

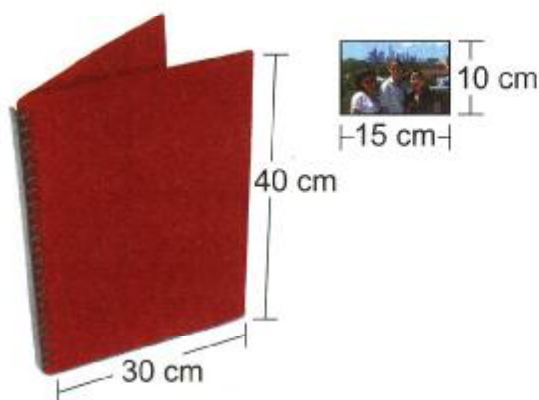
2 heads
4 eyes

In the story there are 12 monsters. Altogether they have 46 eyes.

How many two-headed monsters are in the story?

- (A) 2
- (B) 6
- (C) 10
- (D) 11

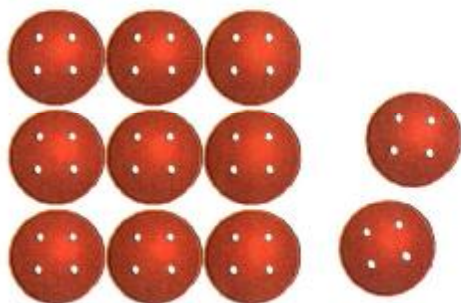
31. Emma wants to stick photos on the outside of her folder. The photos will be placed side by side with no gaps or overlaps.



How many photos will she need to cover just the front **and** the back of her folder?

- (A) 16
- (B) 12
- (C) 8
- (D) 6

32. Lin has 11 buttons. She arranges them to make a square and has 2 buttons left over.

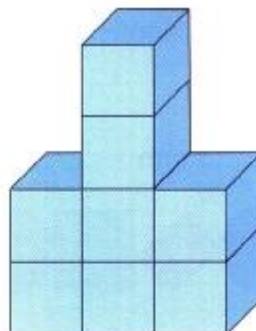


Ahmed has more buttons than Lin. He arranges his buttons to make the biggest square he can. He has 10 buttons left over.

What is the smallest number of buttons Ahmed could have?

- (A) 16
- (B) 26
- (C) 36
- (D) 46

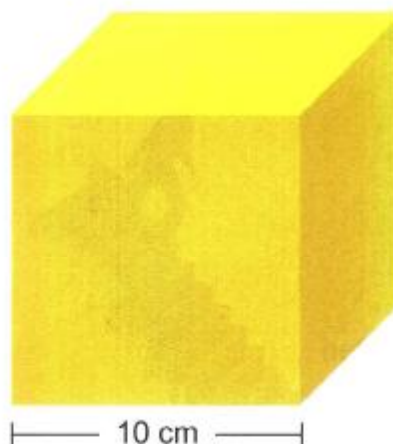
33. This figure is made of small cubes. All surfaces of this figure are painted blue.



How many faces on the small cubes have been painted blue?

- (A) 30
- (B) 32
- (C) 38
- (D) 40

34. Holly has this block of cheese in the shape of a cube.

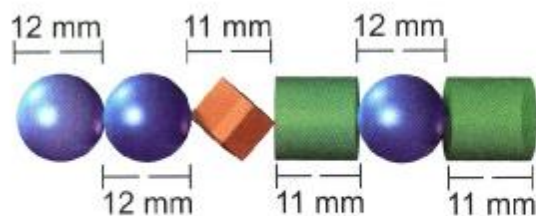


She wants to cut the cheese into smaller cubes each with a side length of 1 cm.

What is the smallest number of cuts she needs to make?

- (A) 18
- (B) 20
- (C) 27
- (D) 30

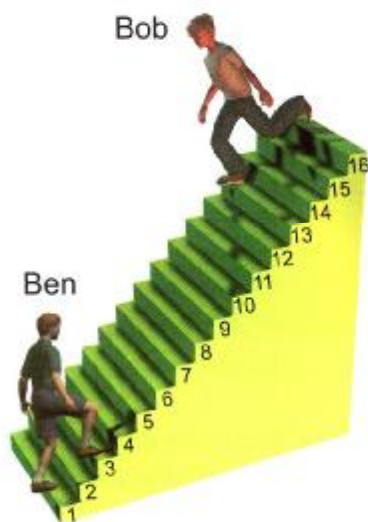
35. Mandy is making a necklace. She has three types of beads: blue spheres, orange cubes and green cylinders. She makes her necklace by repeating this pattern.



How many blue spheres will she need to make a necklace 345 mm long?

- (A) 5
- (B) 10
- (C) 15
- (D) 20

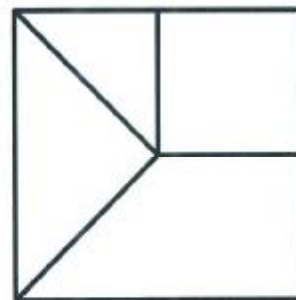
36. Bob walks down the stairs three steps at a time to step 1. Ben walks up the stairs two steps at a time to step 15.



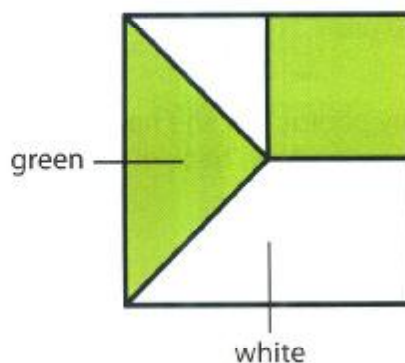
How many steps will **not** be walked on?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

37. This square has four small shapes.



Jacob colours two of the small shapes green. Here is one way he could do this.



In how many different ways can Jacob colour two of the small shapes green?

- (A) 3
- (B) 6
- (C) 8
- (D) 9

38. This truck carries newspapers.



When it is half full of newspapers the truck has a mass of 1762 kg. When it is a quarter full the truck has a mass of 1426 kg.

What is the mass of the truck when it is full of newspapers?

- (A) 1344 kg
- (B) 2434 kg
- (C) 3188 kg
- (D) 3524 kg

39. Jane, Ali and Pat each planted a tree 90 cm in height.

In the time it took Jane's tree to grow 1 cm, Ali's tree grew 2 cm. In the time it took Ali's tree to grow 1 cm, Pat's tree grew 3 cm.

How tall was Jane's tree when Pat's tree was 108 cm in height?

- (A) 18 cm
- (B) 36 cm
- (C) 93 cm
- (D) 96 cm

40. Sandy's music player has five songs displayed.

	Song	Artist
1:	Sum it all up	The Fractions
2:	The difference is clear	L.C. Multiple
3:	Divide and rule	H.C. & The Factors
4:	Square dance	Apple Pi
5:	Product song	Tetra Hedron

Shuffle

When she presses the shuffle button the player puts the songs into a different order. Each song plays only once.

In how many different orders could the songs be played?

- (A) 20
- (B) 25
- (C) 100
- (D) 120

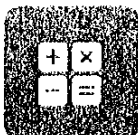
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THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

AUSTRALIA: Year 4
BRUNEI: Primary 4
INDONESIA: Year 5
MALAYSIA: Standard 4
NEW ZEALAND: Year 5
PACIFIC: Year 4
SINGAPORE: Primary 3
SOUTH AFRICA: Grade 4





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2009 Mathematics Answer Keys

ICAS
International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	C	A	B	A	D	C	A	C
2	C	B	D	D	C	B	A	B
3	A	D	B	C	A	D	C	C
4	B	D	D	D	C	C	C	C
5	D	A	D	D	B	D	D	D
6	A	D	A	B	A	A	D	C
7	B	C	A	C	C	A	B	A
8	C	B	D	C	B	C	B	A
9	D	A	C	B	A	B	A	D
10	A	D	C	C	D	C	C	C
11	D	B	B	A	D	D	B	B
12	A	B	D	D	B	B	C	D
13	B	C	B	B	B	C	B	C
14	D	B	D	A	C	B	A	D
15	A	D	C	B	C	B	C	B
16	D	B	C	A	B	A	C	B
17	B	A	D	A	D	C	B	D
18	B	A	B	D	A	B	D	A
19	D	B	A	A	A	D	B	D
20	D	C	B	B	B	C	D	C
21	A	D	A	B	D	A	A	A
22	A	D	A	C	A	A	B	A
23	D	C	D	C	B	D	D	C
24	C	A	A	B	B	B	D	B
25	C	C	B	D	B	B	A	B
26	A	D	B/D	B	C	D	D	D
27	C	C	A	D	D	A	D	D
28	B	A	B	A	A	C	B	C
29	A	B	C	B	C	B	C	A
30	B	C	C	C	B	D	D	A
31	B	A	A	C	D	A	D	B
32	D	B/D	C	A	C	A	B	D
33	C	A	A	C	B	D	A	A
34	A	C	C	D	D	A	A	B
35	C	C	C	B	A	D	C	C
36	D	D	B	18 018	132	116	116	528
37	B	B	D	84 084	18 018	256	324	24 024
38	C	B	B	390	400	24 024	52 052	10 010
39	C	C	C	10 010	24 024	2 02 002	528	76 076
40	D	D	C	361	21 021	970	484	484

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